Lesson 10.3 - One Tailed and Two Tailed Tests

> **Video:** One Tailed and Two Tailed Tests, Critical Values, & Significance Level by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

The big picture: how far is "too far"?

In Lesson 10.1 we wrote H_0 and H_a. In Lesson 10.2 we saw that we might make a wrong decision. This lesson answers the next question: **how far does the sample have to be from H_0 before we reject it?**

1. Start by pretending H_0 is true. If the true mean really is μ_0, sample means still bounce around it by random chance. When we convert the sample mean to a z-score (the **test statistic**, Lesson 10.4), those scores follow the standard normal curve, centered at 0.

2. Near the middle = normal. Far out in a tail = surprising. Most samples land close to 0. A z-score far out in a tail would be very rare **if H_0 were true**. So if we get one, we doubt H_0.

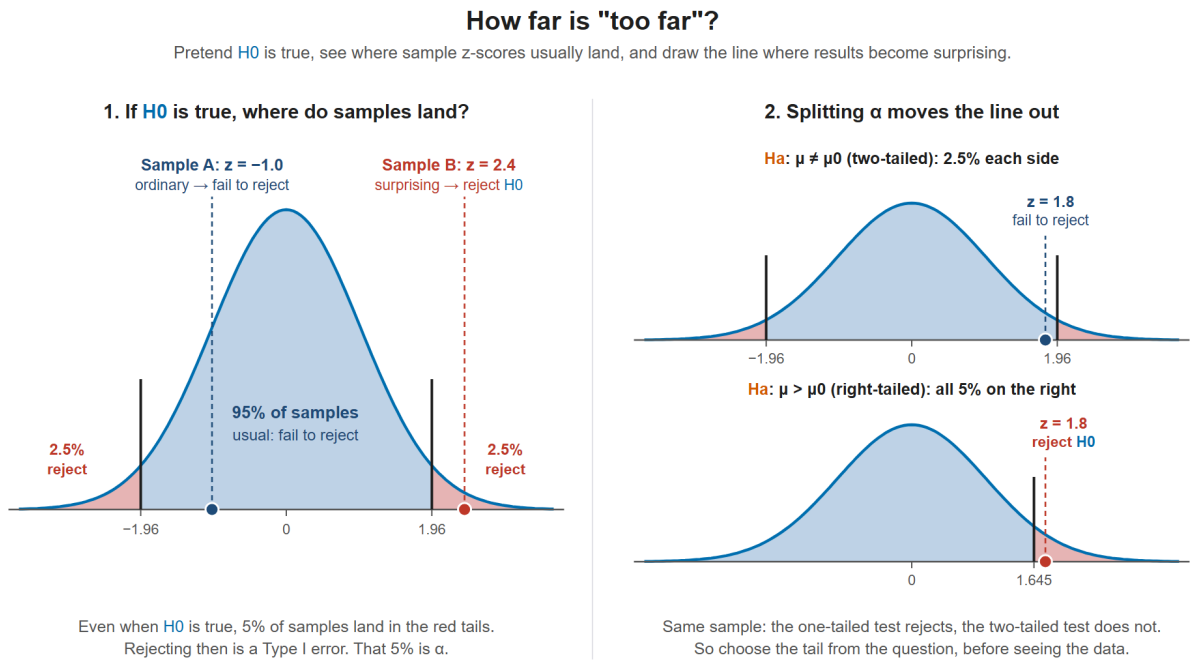
3. Draw a line: the critical value. We need a cutoff for "far enough". That cutoff is the **critical value**, and the tail area beyond it is the **rejection region**.

4. The tail area is α, your false-alarm risk. Even when H_0 is true, a sample lands in the rejection region α of the time (for example 5%). Those are exactly the Type I errors from Lesson 10.2. So **you choose α first**, and it sets where the line goes.

5. Which tail counts depends on H_a. Only results that point **toward H_a** count as evidence.

- Chip bags, $H_a: \mu < 100$ (bags are underfilled). A very **light** sample is evidence. A very heavy sample isn't evidence of underfilling, so only the **left** tail counts.
- $H_a: \mu \neq 100$ (just "different"). Too light **or** too heavy are both evidence, so **both** tails count. The total risk must still be α, so it is **split**: $\alpha/2$ in each tail.

Why the critical values differ. Splitting α pushes each line further out: ± 1.96 for two-tailed, versus 1.645 for one-tailed (both at $\alpha = 0.05$). So a one-tailed test rejects more easily in its chosen direction, but can't detect anything in the other direction.



Left: if H_0 is true, 95% of samples land in the middle. Sample A is ordinary, so we fail to reject. Sample B is out in a tail, so we reject. The red tails still catch 5% of samples even when H_0 is true: that's α, the Type I error rate. Right: the same sample $z = 1.8$ is not far enough for a two-tailed test (line at 1.96) but is far enough for a right-tailed test (line at 1.645).

> **Pick the direction before looking at the data.** It comes from the research question (the wording of H_a), not from which way the sample happened to go. Choosing the tail after seeing the data quietly doubles your real false-alarm risk.

Where this lesson fits in a full test:

1. Write H_0 and H_a (Lesson 10.1).
2. Choose α (often 0.05).
3. **Use the sign in H_a to pick the tail(s) and find the critical value(s).** ← this lesson
4. Compute the test statistic from the sample (Lesson 10.4).
5. Compare: in the rejection region → reject H_0. Otherwise → fail to reject H_0.

Where 1.96 and 1.645 come from. They are z-table lookups. For two-tailed at $\alpha = 0.05$, the area to the left of the cutoff is $1 - 0.025 = 0.975$, so $z = 1.96$. For one-tailed, it is $1 - 0.05 = 0.95$, so $z = 1.645$.

α	Two-tailed critical z	One-tailed critical z
0.10	± 1.645	1.28
0.05	± 1.96	1.645
0.01	± 2.576	2.33

(For a left-tailed test, the one-tailed value is negative: for example -1.645.)

Vocabulary

Term	Meaning
Rejection region	The shaded tail area(s). If the test statistic lands here, reject H_0
Fail-to-reject region	The unshaded middle. If the statistic lands here, keep H_0
Critical value(s)	The z (or t) value(s) separating the two regions
Confidence level c	The area of the fail-to-reject region
Significance level α	The total area of the rejection region: $\alpha = 1 - c$
Calculated value z_c	The **test statistic** computed from the sample (see Lesson 10.4)

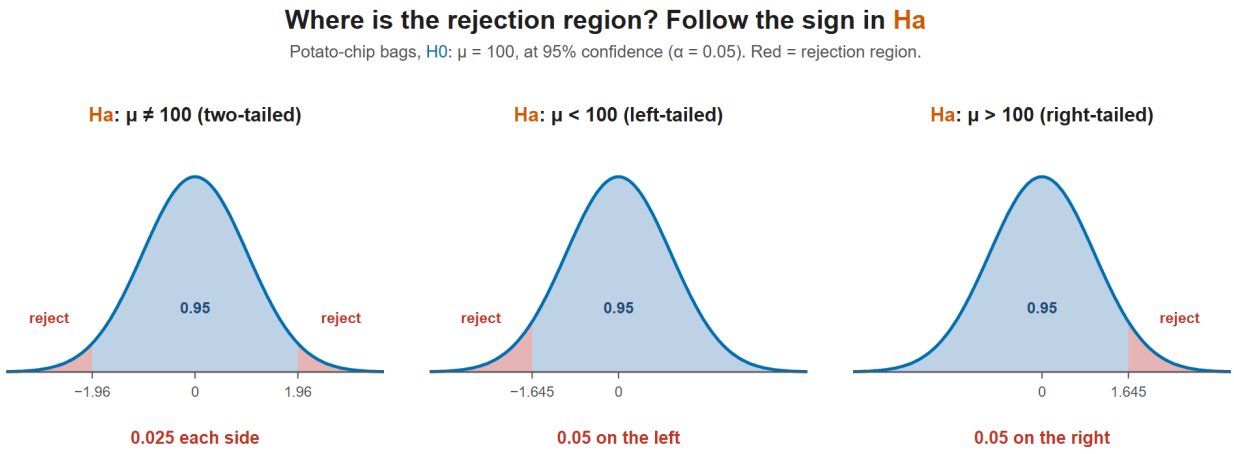
Decision: if z_c falls in the rejection region (beyond the critical value), **reject** H_0. Otherwise, **fail to reject** H_0.

Which test? Read the sign in H_a

H_a	Test	Rejection region	At 95% ($\alpha = 0.05$)
$\mu \neq \mu_0$	**Two-tailed**	Both tails, $\alpha/2$ each	0.025 in each tail, critical $z = \pm 1.96$
$\mu < \mu_0$	**Left-tailed**	Left tail only, area α	0.05 on the left, critical $z = -1.645$
$\mu > \mu_0$	**Right-tailed**	Right tail only, area α	0.05 on the right, critical $z = +1.645$

```
Two-tailed:     ███|                  |███     (α/2 each side)
Left-tailed:    ████|                          (α on the left)
Right-tailed:                         |████    (α on the right)
                      -z       mean       +z
```

The **arrow** in H_a points to the tail: $<$ → left, $>$ → right, $\neq$ → both.



The sign in H_a decides where α goes: split between both tails for ≠, all in the left tail for <, and all in the right tail for >.

Example - potato-chip bags

A company says its chip bags average **100 g**.

- An employee thinks the mean is **not** 100 g: $H_0 : \mu = 100$, $H_a : \mu \neq 100$ → **two-tailed**. At 95% confidence, $\alpha = 0.05$ is split into 0.025 per tail, and 0.95 is in the middle.
- If instead $H_a : \mu < 100$ → **left-tailed**, with all 0.05 in the left tail.
- If instead $H_a : \mu > 100$ → **right-tailed**, with all 0.05 in the right tail.

Key takeaways

- $\alpha = 1 - c$ is the significance level, the size of the rejection region.
- $\neq$ → two tails (split α). $<$ → left tail. $>$ → right tail.
- Compare the calculated statistic with the critical value(s) to decide.